For example, if the computing hardware of a physical host fails, all EC2 instances running on this host will fail. If you're running an application on an affected virtual machine, this application will fail and experience downtime until somebody intervenes by starting a new virtual machine on another physical host. In the case of a small outage—for example, when a single physical host fails—AWS will reboot the affected virtual machines and launch them on a new host. However, in the case of a larger outage—for example, when a whole rack is affected—you are in charge of recovering the failed instances. Therefore, to avoid downtime and to build a highly available system, we recommend enabling autorecovery based on a CloudWatch alarm or distributing the workload among multiple virtual machines.

Examples are 100% covered by the Free Tier

The examples in this chapter are totally covered by the Free Tier. As long as you don't run the examples longer than a few days, you won't pay anything for it. Keep in mind that this applies only if you created a fresh AWS account for this book and there is nothing else going on in your AWS account. Try to complete the chapter within a few days, because you'll clean up your account at the end of the chapter.

High availability describes a system that is operating with almost no downtime. Even if a failure occurs, the system can provide its services most of the time. The Harvard Research Group (HRG) defines high availability with the classification AEC-2, which requires an uptime of 99.99% over a year, or not more than 52 minutes and 35.7 seconds of downtime per year. You can achieve 99.99% uptime with EC2 instances if you follow the instructions in the rest of this chapter. Although a short interruption might be necessary to recover from a failure, no human intervention is needed to instigate the recovery.

High availability vs. fault tolerance